

Phorm

Architecture of a standalone document validation service

Philip Helger, philip@helger.com

phorm 2.2.x, November 2026

Does this document conform?

1,666

registered rule sets

768 active, 898 deprecated

59

rule set groups

Peppol, XRechnung, ZATCA and others

5

REST APIs

below /api

3

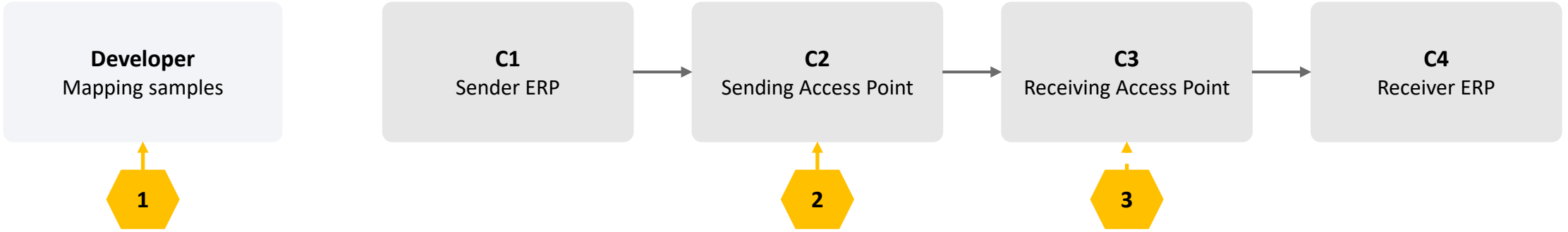
response formats

JSON, XML, HTML

- API only: there is no interactive validation UI
- Stateless: documents are validated and answered, not stored
- A Java WAR on the Photon web framework, deployed to Tomcat
- phive is the engine, phive-rules supplies the rule sets
- ddd detects the document type, kaltblut checks hybrid PDFs
- Java clients use the separate phorm-client library

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Three points in a Peppol exchange



Development time

Validate every sample the mapping produces. No Access Point, network or deployment is needed.

Findings go back into the mapping

C2 outbound gate

Validate before SBDH wrapping and AS4 transmission. Invalid documents never enter the network.

HTTP 400 is the gate decision

C3 on receipt, optional

Validate after AS4 reception, including the SBDH envelope. Phorm only reports the result.

The receiver decides what to do

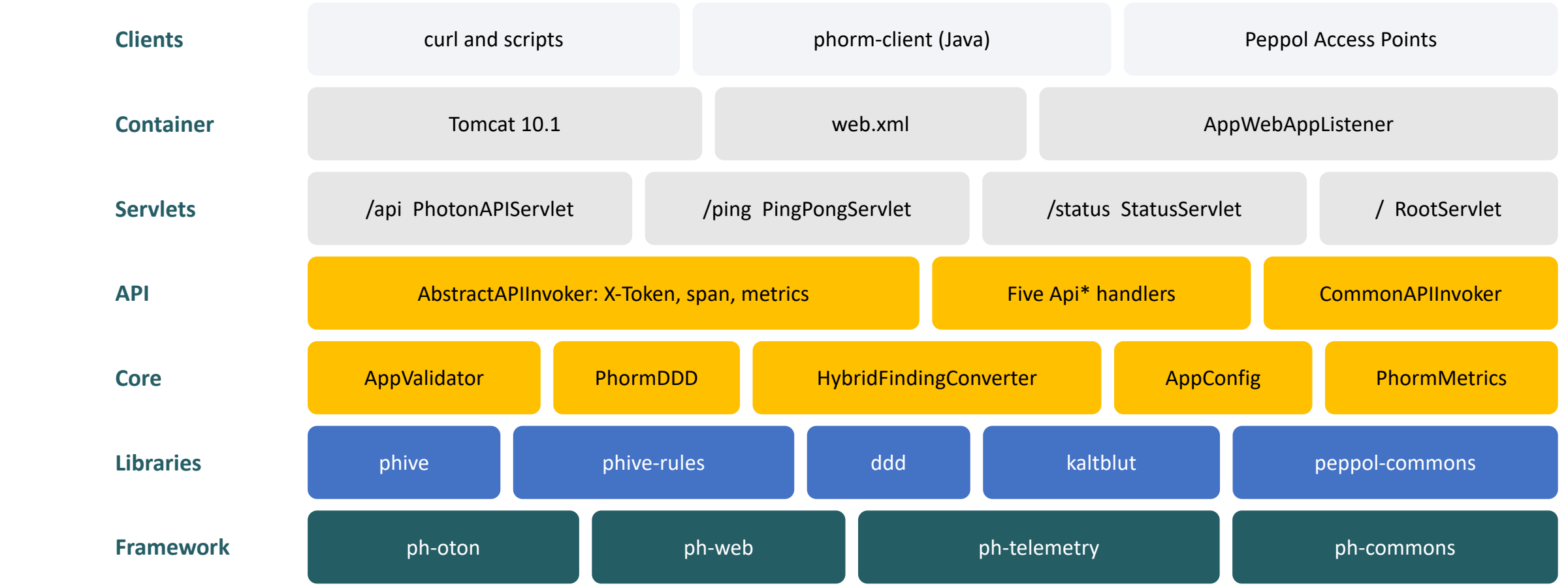
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Architecture

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A thin API layer on libraries



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REST APIs and service endpoints

Method	Path	Handler	Auth	Purpose
POST	`/api/validate/{vesid}`	`ApiPostValidate`	**Token**	Validate against a named rule set
POST	`/api/dd_and_validate`	`ApiPostDetermineDocTypeAndValidate`	**Token**	Detect the type, then validate
POST	`/api/hybrid_validate`	`ApiPostHybridValidate`	**Token**	Validate a hybrid PDF invoice
POST	`/api/determinedoctype`	`ApiPostDetermineDocDetails`	**Token**	Detect the document details
GET	`/api/get/vesids`	`ApiGetAllVESIDs`	None	List the registered rule sets
GET	`/ping`	`PingPongServlet`	None	Health check, answers pong
GET	`/status`	`StatusServlet`	None	Version and status as JSON
GET	`/`	`RootServlet`	None	Landing page

Token: the X-Token header must equal `phorm.api.requiredtoken`. A wrong or missing token gets HTTP 403.

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From raw bytes to rendered findings

Span names carry the prefix phorm.

1	2	3	4	5	6	7
Authenticate	**Read payload**	**Parse XML**	**Determine**	**Resolve**	**Validate**	**Render**
Compare X-Token	Body as bytes	DOM document	Unwrap SBDH or XHE, find the VESID	Look up the rule set	Run phive, add the SBDH layer	JSON, XML or HTML
`auth.check`	`payload.read`	`xml.parse`	`ddd.determine`	`vesid.resolve`	`phive.validate`	`no own span`

403
Token missing or wrong

400
Not XML, no type found, VESID unknown

400
Invalid document, configurable

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How the rule set is selected

Explicit

The caller decides which rules apply.

```
POST /api/validate/{vesid}
```

`eu.peppol.bis3:invoice:latest`

GET /api/get/vesids lists every value.

Automatic, through ddd

1	**Syntax**	Root namespace and local name give the syntax, such as ubl2-invoice
2	**Fields**	XPath expressions read customization, process, sender and receiver
3	**Value provider**	The Customization ID selects the profile and its VESID
4	**Resolution**	The VESID is parsed and looked up in the registry

AppValidator holds one registry filled from phive-rules-all and phive-rules-all-legacy, loaded on first use of the class

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Hybrid PDF invoices



All findings land in one list, carrier layers first



No embedded XML?

Phorm returns the carrier results only

Unparsable PDF?

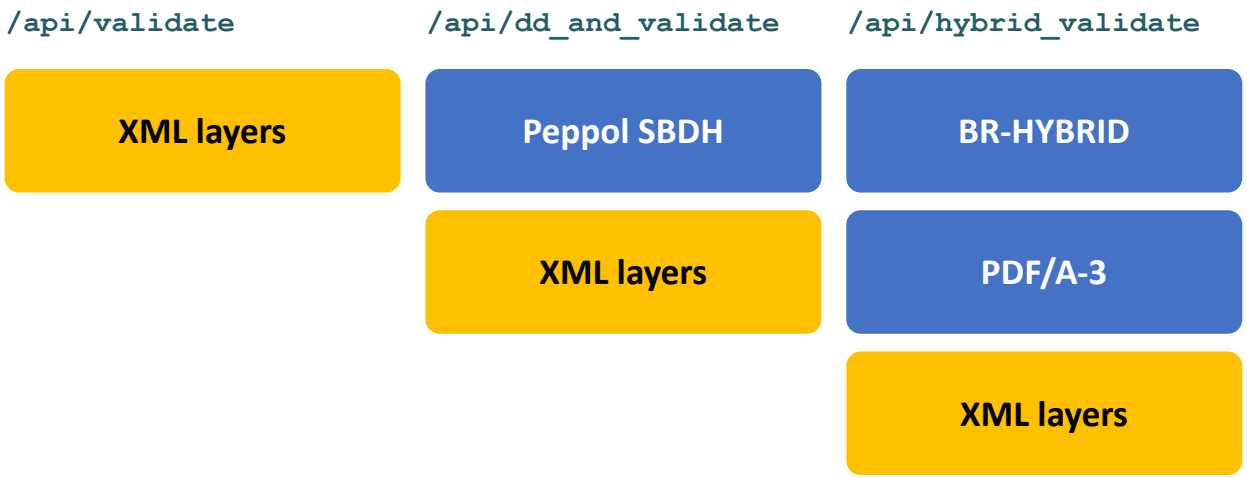
Validation stops, no XML rules are run

country=DE|FR|OTHER

Selects the country specific BR-HYBRID rules

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One ValidationResultList



Layers in result order. The SBDH layer appears only for wrapped documents; for Peppol BIS Billing and PINT it also cross-checks sender and receiver IDs.

Accept header	Response
application/xml	XML
text/html	HTML report
anything else	JSON

Outcome	HTTP status
Valid document	200
Invalid document	400, or 200 if configured
Unusable request	400
Missing or wrong token	403

The status alone can act as a gate. Set `phorm.api.response.onfailure.http400` to false to keep 200.

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Startup and configuration

AppWebAppListener

1. **`initGlobalSettings`**
   Logging, request tracker, token check, telemetry
2. **`initLocales`**
   Register the default locale
3. **`initAPI`**
   Register the five API descriptors
4. **`initSecurity`**
   Audit disabled, predefined users
5. **`initManagers`**
   Error handler and meta manager

Lookup order, first match wins

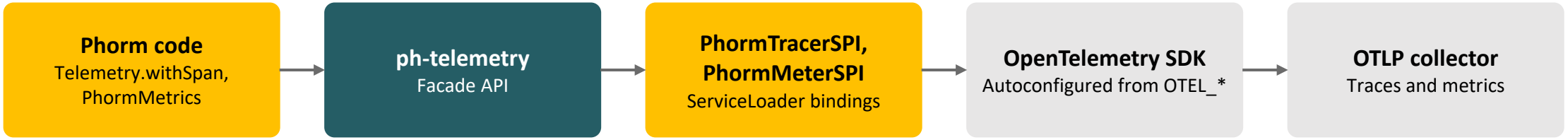


Key settings

`phorm.api.requiredtoken`	Mandatory API token
`phorm.api.response.onfailure.http400`	Default true
`phorm.api.response.log.payload`	Default false
`phorm.telemetry.enabled`	Default false
`webapp.datapath`	Data folder

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Telemetry is opt-in



Spans

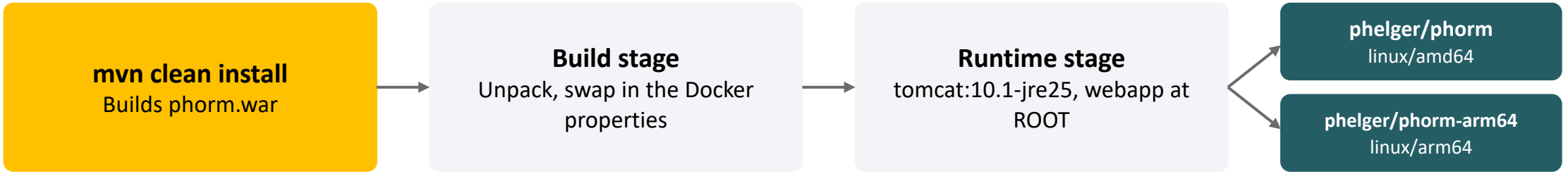
```
phorm.api.{endpoint}
    phorm.auth.check
    phorm.payload.read
    phorm.xml.parse
    phorm.ddd.determine
    phorm.vesid.resolve
    phorm.phive.validate
    phorm.kaltblut.validate
    phorm.kaltblut.extract_xml
```

Metrics

```
phorm.requests.received
phorm.validation.runs
phorm.validation.duration
phorm.validation.findings
phorm.payload.bytes
phorm.ddd.determinations
phorm.vesid.resolutions
phorm.kaltblut.runs
phorm.kaltblut.embedded_xml.bytes
phorm.ves.registry.size
```


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One WAR, two container images



Run

docker run -p 8080:8080, data in /phorm/data

Health

/ping for liveness, /status for version details

Hardening

The image blocks 16 compromised eMudhra CA certificates

Before go-live

Replace the default phorm-dev-token value

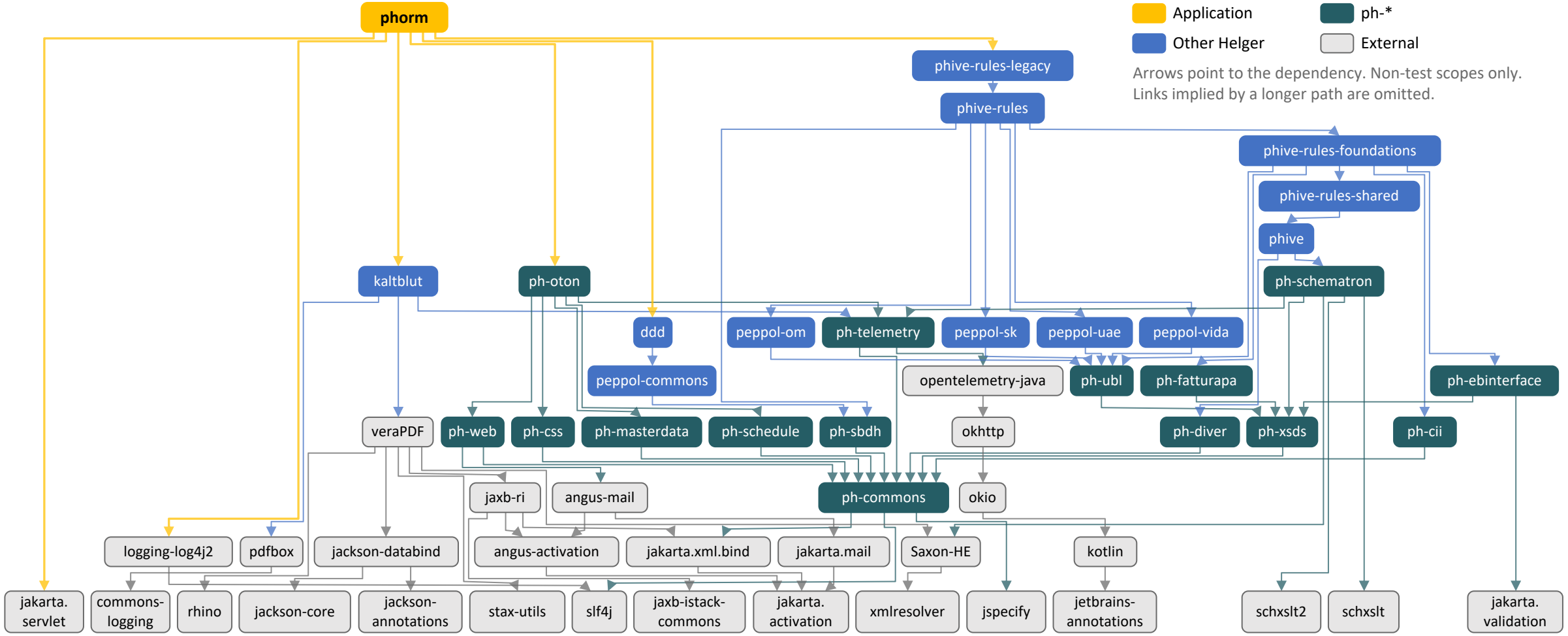
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Dependencies

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Repository dependencies



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What the dependency graph shows

28

Helger repositories

15 ph-*, 12 others, plus phorm

29

external repositories

Apache, Jakarta, OpenTelemetry and others

15

leaf repositories

All external, all on the bottom row

1

ph-commons is the hub

Every other Helger repository in the graph depends on it, directly or indirectly.

2

The rule chain is the longest path

phive-rules-legacy, phive-rules, phive-rules-foundations and phive-rules-shared stack up on phive.

3

Telemetry adds depth, not coupling

ph-telemetry pulls OpenTelemetry, OkHttp and Kotlin into the graph, but stays a no-op until enabled.

4

PDF is the external-heavy branch

kaltblut brings in veraPDF and PDFBox, and through them Jackson, Rhino and the JAXB runtime.

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Summary



- Phorm keeps little logic of its own
- One stable HTTP contract over phive, the rule sets, ddd and kaltblut
- API only, with no interactive UI
- One result model for every check
- The same rules at design time, at C2 and at C3

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